INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

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THE SITE

Location	TierPoint Sioux Falls West Data Center, SD -- station KFSD, America/Chicago
Building	OSM 239032142 TierPoint Sioux Falls West Data Center
Facade gap	not applicable -- one building, no second facade
Weather record	43,797 real hourly records from KFSD
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-12-05 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 10 of 24 hours with 1 mode change. The reactive incumbent took 13 hours -- 3 more -- but declared 1 unsafe hour against the agent's 0. 12 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	10 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	13 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) 12 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

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hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	0.810	18.0	0.560	1.133
01	mechanical	yes	switch budget	1.110	18.0	1.110	1.053
02	mechanical	yes	switch budget	0.840	18.0	0.560	0.990
03	mechanical	yes	switch budget	0.840	18.0	0.560	0.915
04	mechanical	yes	switch budget	1.390	18.0	1.110	0.865
05	mechanical	yes	switch budget	0.585	18.0	0.000	0.813
06	mechanical	yes	switch budget	0.530	18.0	-0.610	0.862
07	mechanical	yes	switch budget	1.915	18.0	-0.610	1.430
08	mechanical	yes	switch budget	3.335	18.0	0.560	1.926
09	mechanical	yes	switch budget	4.695	18.0	1.670	2.219
10	mechanical	no	dry-bulb	25.250	18.0	42.220	2.091
11	mechanical	yes	-	7.500	18.0	6.110	1.941
12	mechanical	yes	-	7.500	18.0	6.110	1.729
13	mechanical	no	dry-bulb	26.945	18.0	6.110	1.317
14	FREE	yes	-	7.500	18.0	4.440	1.161
15	FREE	yes	-	5.830	18.0	2.220	1.047

16	FREE	yes	-	4.445	18.0	0.560	1.026
17	FREE	yes	-	2.470	18.0	-0.610	1.042
18	FREE	yes	-	0.830	18.0	-1.110	1.290
19	FREE	yes	-	-0.525	18.0	-1.110	1.161
20	FREE	yes	-	-1.415	18.0	-2.220	1.654
21	FREE	yes	-	-1.665	18.0	-2.220	1.606
22	FREE	yes	-	-1.665	18.0	-2.220	1.363
23	FREE	yes	-	-2.220	18.0	-2.220	1.059

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.810 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.110 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.840 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.840 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.390 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.585 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.530 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.915 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.335 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.695 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.250 C against a 18.0 C limit, so it fails by 7.250 C. It would take a limit 7.250 C higher, or a bound 7.250 C tighter, to change this hour.

11:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.500 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.500 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.945 C against a 18.0 C limit, so it fails by 8.945 C. It would take a limit 8.945 C higher, or a bound 8.945 C tighter, to change this hour.

14:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.500 C, which is 10.500 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.161 C, and the margin is measured, not chosen: 1.161 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

15:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.830 C, which is 12.170 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.047 C, and the margin is measured, not chosen: 1.047 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 4.445 C, which is 13.555 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.026 C, and the margin is measured, not chosen: 1.026 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 2.470 C, which is 15.530 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.042 C, and the margin is measured, not chosen: 1.042 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.830 C, which is 17.170 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.290 C, and the margin is measured, not chosen: 1.290 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.525 C, which is 18.525 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.161 C, and the margin is measured, not chosen: 1.161 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -1.415 C, which is 19.415 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.654 C, and the margin is measured, not chosen: 1.654 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -1.665 C, which is 19.665 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.606 C, and the margin is measured, not chosen: 1.606 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -1.665 C, which is 19.665 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.363 C, and the margin is measured, not chosen: 1.363 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -2.220 C, which is 20.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.059 C, and the margin is measured, not chosen: 1.059 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

WHAT IT IS WORTH OVER FIVE REAL YEARS, AT THIS SITE

Held-out days simulated	913 -- the agent never calibrates on a day it is scored on
Free cooling delivered	17.09 h/day, i.e. 6,242 h/yr
Chiller-hours avoided	+354.8 h/yr against a tuned reactive incumbent
12-hour plan stability	93.7 % of 21,901 re-plans change nothing at all

The ladder, one constraint at a time:

N-56-like: notice 0, skill 1.00, no constraint	+37.2 h/yr
+ switch budget 2, min dwell 3 h	+42.8 h/yr
+ dew-point gate 15 C (Green Grid WP#46 p.6)	+88.4 h/yr
+ notice 3 h, skill 0.50 (no perfect forecast)	+354.8 h/yr
+ unanchored, 4 measured FG offsets rotated	-159.2 h/yr

PRICED, IN THIS STATE'S OWN ELECTRICITY TARIFF

Chiller power	156.1 kW per MW of IT load (0.549 kW/ton, the ASHRAE 90.1-2019 minimum)
Electricity	8.72 cents/kWh (Virginia commercial, 2024 annual)
Energy avoided	55,394 kWh per MW of IT load per year
Value	$4,830 per MW of IT load per year

Everything above is PER MEGAWATT OF IT LOAD. This project has never measured a data centre's size and will not invent one; a reader who knows their own IT load multiplies once.

WHAT IS NOT CLAIMED

- The chiller COMPRESSOR only. Fans, chilled-water pumps, condenser pumps and cooling-tower fans keep running, and an airside economizer moves MORE air, so fan power can rise. The unmeasured term has the OPPOSITE SIGN, so this is an upper bound.
- Code-minimum chiller efficiency is the OPTIMISTIC end: Standard 90.1 is a floor, hyperscale plants beat it, and a better chiller saves less per hour switched off.
- Full-load kW/ton OVERSTATES the draw at the moment free cooling is available, because free cooling happens when it is cool and a chiller at a cool condenser runs below its rating. IPLV is the lower published number but is an annual-weighted metric under AHRI's assumed load profile, not the part-load point free-cooling weather sits at.
- The heat rejected is APPROXIMATED by the IT load. UPS, PDU and lighting losses inside the envelope add to it; some overhead sits outside the cooled space, so scaling by full PUE would overshoot. No PUE multiplier is applied.

- EIA STATE-SECTOR AVERAGES, not the site's tariff. A Loudoun County campus buys on a large-general-service contract that is not public.
- No carbon, no water, no maintenance and no demand charges. Tower water use moves the OTHER way when the chiller runs less.
- Every caveat on the hours is inherited: the headline is conditional on the bank sitting on the long facade (the refusal guard is priced at -3,124 h/yr where it fires) and the unanchored row is NEGATIVE. Those rows appear here with their signs intact.

HOW TO REPRODUCE EVERY NUMBER IN THIS REPORT

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cd INTAKE-ARBITER/src && python run_all.py
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That rebuilds every artefact from saved data and then audits it: 39 checks, 68 published figures re-read out of the files the code itself wrote, and it exits non-zero on any failure. It makes ZERO API calls -- every input is a saved FortyGuard response, a committed geometry file, or the station's own hourly record.

Generated by INTAKE-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.